

SECTION-1 VRF UNIT (HEAT PUMP TYPE)

1.1 VRF Outdoor Unit

- i. The outdoor unit shall be factory assembled, weather proof casing (Material of construction of casing shall be vendor's standard design), constructed from heavy gauge GI sheets steel panels and coated with baked enamel finish. The outdoor unit shall be completely factory wired, tested with all necessary controls & filled with first charge of refrigerant before delivering at site.
- ii. The inverter technology based D.C Twin Rotary / Scroll compressor modular type VRV/ VRF equipment (Heat pump type suitable for cooling as well as heating) should of designed, so that refrigerant piping between outdoor units and furthest indoor unit shall be extendable up to 225m. Allowable level difference between outdoor & indoor unit shall be 50m in case of outdoor unit on top & 40m in case of outdoor unit at bottom. Allowable level difference between two indoor units connected to same outdoor unit shall be up to 15m. All the outdoor units comprising of multiple modules should have 100% inverter type compressor in each module.
- iii. For improvement in efficiency other than ISEER value, the units should comply with minimum COP of 3.7 at 100% load in cooling mode at AHRI condition and each module should have all inverter compressor/unit.
- iv. The outdoor units shall be suitable to operate within an ambient temperature range of 5 Deg C to 52 Deg C in cooling mode and -20 Deg C to 15 Deg C in heating mode.
- v. The above COP values as indicated are required to be furnished in Original by the tenderer directly from the original equipment Manufacturer (OEM) with OEM's seal and signature on all documents along with relevant pages from Engineering Handbook.
- vi. The outdoor unit shall be factory tested and filled with first charge of refrigerant R-410A before delivering at site.
- vii. It should also be provided with duty cycling for D.C inverter Twin Rotary/ Scroll compressors capable of changing the rotating speed of compressor by inverter controller to follow variation in cooling & heating loads & switching starting sequence for better stability and prolonging equipment life or similar features if available in D.C Twin Rotary / Scroll will also be accepted.
- viii. The unit shall be provided with its own microprocessor control panel with provision for integration with the building management system for Air-conditioning system.
- ix. The machine must have a sub cool feature to use coil surface more effectively through proper circuit/ bridge so that it prevents the flushing of refrigerant from long piping due to this effect thereby achieving energy savings.
- x. The outdoor unit should be fitted with low noise level and should not be more than 67 db (A) at normal operation when measured at a point 1 mtr.

- In front of the unit at a height of 1.5 mtrs.
- xi. The outdoor unit should be fitted with low noise aero spiral design fan with aero fitting grill for spiral discharge airflow to reduce pressure loss and should be fixed with DC/AC fan motor for better efficiency.
 - xii. In case of trouble occurs in an indoor units (s), the continuous operation of system should be possible after taking corrective bypass action.
 - xiii. The unit shall be designed in such as way that cleaning of drain Pan should be easy & inspection/ replacement of compressor should be easy.
 - xiv. The condensing unit shall be designed to operate safely when connected to multiple fan coil units.
 - xv. The outdoor unit shall have refrigerant/ air cool PCB chamber for better operation at high ambient temperature.
 - xvi. The outdoor unit shall have feature to change the evaporative temperature with respect to load for better comfort and energy efficiency.
 - xvii. The systems shall have free phase technology & operation shall be continuous in case of phase reverse in supply electricity.
 - xviii. The system shall have automatic/manual refrigerant charge function for optimal charging for additional refrigerant.
 - xix. The system shall deliver same capacity @ 39deg C ambient temperature without deration.
 - xx. The fan static pressure of the outdoor unit shall be minimum 75 Pa to avoid hot air recirculation.
 - xxi. Top discharge ODU shall be provided with cowl complete with grill and bird screen to dispatch heat and to avoid making hot pocket in balcony.
- The cowl should be painted to match with the ODU's color.

1.2 Compressor

- a. The compressor is inverter based D.C Twin Rotary / Scroll compressor system shall be highly efficient. The system should response efficiently in accordance to the variation in cooling or heating load requirement.
- b. All outdoor units shall have multiple steps of capacity control to meet load fluctuation and indoor unit individual control. All parts of compressor shall be sufficiently lubricated stock. Forced lubrication may also be employed.
- c. Oil heaters shall be provided in the compressor casing.

1.2.1 Oil Recovery system

- a. Unit shall be equipped with an oil recovery system to ensure stable operation with long refrigeration piping lengths. The system shall have oil recovery cycle of 8 or more hours.
- b. The system must be provided with oil balancing circuit to avoid poor lubrication.

1.2.2 Refrigerant Circuit

- a. The refrigerant circuit shall include liquid and gas shut-off valves and a solenoid valves at condenser end.
- b. The equipment must have inbuilt refrigerant stabilization control for proper refrigerant distribution.
- c. All necessary safety devices shall be provided to ensure the safe operation of the system.

1.2.3 Heat Exchanger

- a. The heat exchanger shall be constructed with copper tubes mechanically bonded to aluminium fins to form a cross fin coil.
- b. The aluminium fins shall be covered by anti-corrosion resin film.
- c. The unit shall be provided with necessary number of direct driven low noise level propeller type fans arranged for vertical discharge. Each fan shall have a safety guard.

1.2.4 Safety Devices

- a. All necessary safety devices shall be provided to ensure safe operation of the system.
- b. Whatever safety devices are required shall be part of outdoor unit:- high pressure switch, fuse, fan drive overload protector, fusible plug, crankcase heater, over load relay, overload protection for inverter based technology.
- c. The outdoor roof mounted units shall be provided in such a way that these do not affect the overall aesthetics and ambience of the building. If required, these units shall be suitably camouflaged to give good aesthetic look.
- d. Noise levels for outdoor units shall not be more than 67 db (measured at a point 1 meter in front of the unit at a height of 1.5 meters).

1.3 Indoor Units

All indoor units as specified shall have:-

- i) Each unit shall have electronic control valve to control refrigerant flow rate respond to load variation of the room.
- ii) The address of the indoor unit shall be set automatically or through central controller in case of individual and group control.
- iii) The fan shall be dual suction, aerodynamically designed, Turbo, multi blade type, statically & dynamically balanced to ensure low noise and vibration free operation of the system. The fan shall be direct driven type, mounted directly on motor shaft having support from housing.
- iv) The cooling coil shall be made out of seamless copper tubes and having continuous aluminium fins. The fins shall be spaced by collars forming an integral part. The tubes shall be staggered in the direction of airflow. The tubes shall be hydraulically/ mechanically expanded for minimum thermal contact resistance with fins. Each coil shall be factory tested at 21 kg/sq.cm air pressure under water.
- v) Indoor unit shall have cleanable type filter fixed to an integrally moduled / moduled plastic frame. The filter shall be slide in and neatly insert able type. It shall be possible to clean the filters either with compressed air or water.
- vi) Each indoor unit shall be with corded/ cordless remote controller as standard features. The controller shall be able to change fan speed.
- vii) All indoor Units should have force shut option to connect with fire alarm adaptor, so that the unit can be integrated with other systems to stop automatically such as in case of fire.

1.3.1 Ceiling mounted Cassette Type Unit-4way/Round Flow with sensing

The unit shall be ceiling mounted cassette type. The unit shall include pre-filters, fan section and DX-coil section, high lift drain pump, and Low gas detection system. The plastic body shall be light in weight and shall be able to suspend. The fan shall be aerodynamically designed diffuser turbo fan type. Unit shall have an external attractive panel for supply and return air.

1.3.2 Ceiling Mounted Cassette Unit - One Way Type

The unit shall be ceiling mounted corner cassette type. The unit shall include prefilter, fan section and DX coil section, high lift drain pump, and Low gas detection system. The housing of the unit shall be powder coated galvanized steel. The body shall be light in weight and shall be able to suspend from four corners. Unit shall have an external attractive panel for supply and return air. Each unit shall have high lift drain pump, and Low gas detection system.

1.3.3 Ceiling Mounted Cassette Unit - Two Way Type

The unit shall be ceiling mounted airflow type cassette unit. The unit shall include pre-filter, fan section and DX coil section, drain pump. The housing of the unit shall be powder coated galvanized steel. The body shall be light in weight and shall be able to suspend from four corners. Unit shall have an external attractive panel for supply and return air. Each unit shall have high lift drain pump, and Low gas detection system.

1.3.4 Hi-wall Type Indoor Unit

High wall type Indoor unit shall be equipped with washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, drain pan, necessary accessories etc., suitable for operation on 230 V $\pm$ 10%, 50 Hz, single phase AC supply.

1.3.5 Ceiling Mounted Ductable Type Unit

Unit shall be suitable for ceiling mounted type. The unit shall include pre filter, fan section & DX coil section and should have drain pump.

1.3.6 Floor Standing Duct Type:

The unit shall include pre filter, fan section & DX coil section. Floor standing units shall have High Static Pressure and Belt drive system that allows for use of air discharge outlet in various shapes as well as long duct. The minimum External static pressure of unit shall be 400 Pa. The housing of unit shall be light weight powder coated galvanized steel.

1.3.7 Heat Recovery Ventilation Unit

Unit shall be suitable for ceiling mounted type. The unit shall include pre filter, heat exchange element, fan section supply air fan and Exhaust air fan and multispeed motor and wired remote.

1.3.8 Treated Fresh Air Unit

Unit shall be suitable for ceiling mounted type. The unit shall include pre filter, fan section & DX coil section. The units should be capable to handle 100% fresh air.

1.4 PC based central controller

The contractor has to provide PC based central controller for controlling (ON/OFF, scheduling, lock/unlock, swing control, temperature setting, error detection, filter detection) all indoor units (256) add monitoring of corresponding outdoor units from one central location. PC based central controller should be BMS (like Bacnet, Lan works, Modbus) compatible. It must show the status of all indoor units along with outdoor unit.

1.5 Refnet Joints/ Header Joints

Supply & installation of the Y-Joint/Ref-net separation refrigeration pipe joints and header in the appropriate orientation to enable correct distribution of refrigerant. The Distribution Joints should be factory insulated with close cell nitrile rubber tubular sections of class O fire ratting.

1.6 Refrigerant Piping

- i) Refrigerant piping for the air-conditioning system shall be upto 19 mm dia of soft seamless copper tubes & for above 19.1 mm dia the pipe material shall be of hard seamless copper tubes with pipes material being hard drawn copper pipe. Forged copper fittings shall be used for the refrigerant piping. The refrigerant piping arrangements shall be in accordance with good engineering practices as applicable to the air-conditioning industry, and shall include charging connections, suction line insulation and all other items normally forming part of proper refrigerant circuits except Y joint/separation tubes.
- ii) Before joining any copper pipe or fittings, its internals shall be thoroughly cleaned by passing a clean cloth via wire or cable through its entire length. The piping shall be continuously kept clean of dirt etc. while constructing the joints. Subsequently it shall be thoroughly blown out using nitrogen gas.
- iii) After completion of installation of the refrigerant piping, the refrigerant piping system shall be pressure tested using nitrogen gas at a suitable pressure as specified by OEM (Original Equipment Manufacturer). Pressure shall be maintained in the system for 48 hours. The system shall then be evacuated to vacuum of not less than 700 mm Hg and held for 24 hours.
- iv) The sizes as designed and erect proper interconnections of the complete refrigerant circuit. The thickness of copper piping shall as per manufacturer recommendations.
- v) The suction line pipe size and the liquid line pipe sizes shall be selected according to the manufacturer's specified diameter. All refrigerant pipes shall be properly supported and anchored to the building/structure using steel hangers, fasteners, brackets and supports which shall be fixed to the building/structure by means of inserts or expansion shields or anchor fasteners of adequate size and number to support the load imposed thereon.

- vi) The refrigerant piping should be laid in such a way that it should not distort the interior of the room, wherever the refrigerant pipe has to be laid across the room, it should be laid in a concealed manner by making appropriate boxing arrangement with minimum 15 mm thick Gypsum Board matching with the interior of the room. All associated minor Civil Engineering works (like chasing on wall, ceiling & re-plastering & repainting etc.) related with the above items are included in the scope of work. The above scope does not include false ceiling wherever required.
- vii) The refrigerant pipes shall be insulated with Nitrile. To protect Nitrile rubber insulation of outdoor installed copper piping from degradation due to ultra violet rays and atmospheric condition, it shall be covered with polyshield coating of at least two coats of resin and hardener (poly bond make or equivalent). Fiber glass tape shall be helically wound with adequate overlap & coated with two coats of resin with hardener to give smooth & plain finish.
- viii) Entire liquid and suction refrigerant pipe lines including all fittings, valves and strainer bodies, etc. shall be insulated with 19mm/ 13mm thick electrometric nitrile rubber as required.

1.7 Drain Piping duly insulated

- i) The drain pipe connection of each cassette unit to the main header should be 32mm dia and the header pipe should be of 40mm dia. The drain-pipe should be heavy duty PVC pipe ISI marked and conforming to relevant IS complete with fitting as required whereas the connection of the fan coil unit to the PVC pipe should be with U trap & flexible braided pipe. The drain piping should be insulated with 6 mm thick tubular nitrile rubber insulation.
- ii) For proper drainage of condensate U trap shall be provided in the drain piping wherever required. All pipe supports shall be prefabricated and pre-painted slotted angle support, properly installed with clamps. The condensate drain pipe arrangement for disposal of condensate water be made in such a way that there should not be any leakages of condensate water inside rooms as well in the route of drain water pipe line & water should be discharged at the location jointly decided with Engineer-in-Charge of work. All associated Civil Engineering works as per requirement at site in above connection like making chase in the wall & restoring it original shape by re-plastering & repainting, etc. are included in the scope of work. The arrangement of drain-pipe shall be made in such a way that it should not affect the aesthetic of the building as well as is maintenance friendly & easily accessible.

1.8 Rigid PVC Conduit Pipe

- i) Laying conduit shall be in conformity with IS: 732/1989 (Latest Version), IS 4648/1968 (Latest Version).

- ii) Fitting for rigid non-metallic conduit shall conform to IS 3419/1989 (Latest Version) and accessories shall conform to relevant IS.

1.9 Various Sizes PVC Insulated Copper Conductor Wiring Cables

PVC insulated multi stranded sheathed copper conductor wiring cable for working voltage upto & including 1100 Volts, ISI marked conforming to IS 694/1990 (Latest Version).

Wiring of installation shall be in conformity with IS 732/1989 (Latest Version), IS 4648/1968 (Latest Version).

SECTION-2 SPLIT AIR CONDITIONING UNITS

1. Scope

The scope of this section comprise the supply, erection, testing and commissioning of Air Cooled Split Units conforming to these specifications.

2. Type

The Split Units (Cassette / Hi-Wall) shall be BEE 5 star rating (applicable as on date of supply of item) consist of Inverter compressor, motor, air cooled condenser, integral refrigerant piping and wiring, all mounted on a steel frame.

3. Indoor unit to be installed for Split Unit within building, shall be housed in insulated cabinet consisting of cooling coil, blower with motor, filter & insulated drain pan. Split unit must deliver specified capacity after taking into account losses due to piping length & site conditions.

4. Capacity

The refrigeration capacity of split unit shall be as per actual requirement.

5. Compressor and Motor

Compressor shall be inverter type and shall have dual pressure stat, and an operating oil charge. The motor shall be suction gas cooled and shall be sealed against dirt and moisture. The motor shall be suitable for 220±10 % volts, 50 Hz, A.C. supply.

6. Refrigerant Piping and Controls

Refrigerant piping and fittings interconnecting compressor condenser shall be all copper and valves shall be brass / gunmetal construction. The refrigerant used shall be ozone friendly HFC R410a/R-32/R-134a/R-407c.

7. Casing

The indoor & outdoor units shall be sectionalized / cabinet construction. Indoor units shall be consisting of fan section, coil section, filter section, and drain pan. Outdoor unit shall consist of condenser coil, fan & compressor. The compressor shall be mounted with the outdoor units. Each section shall be constructed of thick sheet steel all welded / bolted construction, adequately reinforced with structural members and provided with sufficient access panels for proper lubrication and maintenance. Base panel shall be constructed of fabricated steel structure provided with an under frame suitably braced. Drain pan shall extend under coil and fan sections with drain

connections. Removable panels in fan and coil sections shall provide access to all internal parts.

8. Fan Motor and Drive

Fan motor shall be suitable for 220±10% volts, 50 Hz, A.C. Supply, Single phase, motors shall be provided with permanent capacitor. Motors shall be especially designed for quite operation.

9. Fan

Fan wheels and housing shall be fabricated from heavy gauge steel. Fan wheels shall be of double-width, double inlet forward-curve, multi-blade type enclosed in a housing and mounted on a common shaft. Fan housing shall be made of die-formed steel sheets with stream-lined inlets to ensure smooth air flow into the fans, fan shaft bearing shall be oil/grease lubricated. All rotating parts shall be dynamically balanced individually, and the complete assembly shall be statically and hydraulically balanced.

10. Cooling Coil (As per OEM)

Cooling coils shall be of fin and tube type having aluminum fins firmly bonded to copper tubes assembled in zinc coated steel frame. Face and surface areas shall be such as to ensure rated capacity from each unit and air velocity across each coil shall not exceed 100 meters per minute. The coil shall be pitched in the unit casing for proper drainage. Each coil shall be factory-tested at 21 Kg. per sq.cm air pressure under water. Water coils shall be designed for a maximum working pressure of 10 kg/sq.cm. Tube shall be mechanically/hydraulically expanded for minimum thermal contract resistance with fins. The number of fins per cm. shall be 4 to 5.

11. Vibration Isolators

The indoor and outdoor units shall be provided with ribbed rubber pad vibration isolators.

12. Painting

Shop coats of paint that have become marred during shipment or erection shall be cleaned off with mineral spirits, then coated with enamel paint to match the finish over the adjoining shop-painted surface.

13. Performance Rating

The unit shall be selected for the lowest operating noise level.

SECTION-3 AIR HANDLING UNIT

1. Floor Mounted/Ceiling Suspended DX Air Handling Units

The air handling units shall be double skin construction, comprising of various sections such as mixing box, filter section, coil section and fan section as per designed requirement.

2. Casing:

Double skinned panel to be fabricated with anodized extruded aluminum section framework bolted together with sandwich panel having 22 gauge pre-coated outer skin and 22 gauge plain GI inner skins injected with 25/43mm thick PU foam insulation having density of $38 \pm 2 \text{ kg/m}^3$. The entire frame shall be mounted on a sheet steel channel base of 100 mm height. The panels shall be sealed to the framework by heavy duty 'O' ring gaskets held captive in the framed extrusion. All panels shall be detachable or hinged. Hinges shall be made of die cast aluminum / engg. nylon plastic with stainless steel pivots, handles shall be made of hard nylon and be operational from both inside and outside of the unit. Units supplied with various sections shall be suitable for onsite assembly with continuous foam gasket. All fixing and gaskets shall be concealed.

Units shall have hinged quick opening access doors in the fan and filter section. Access doors shall also be double skin type.

Condensate drain pan shall be fabricated from 18 gauge stainless steel sheet. It shall be isolated from bottom floor panel through PU foam insulation.

Outdoor AHU shall be with thermal break profile casing.

a. Motor (VFD Compatible)and Drive:

Fan motors shall be IE-3 efficiency class and shall be $415 \pm 10\%$ volts, 50cycles, three phases, totally enclosed fan-cooled class F, with IP-55 protection. Motors shall be especially designed for quiet operation and motor speed shall not exceed 1440 rpm. Drive to fan shall be provided through 'V' belt-drive arrangement. Belts shall be of the oil-resistant type. Minimum efficiency of shall not be below 70%.

b. Fan:

Fans shall be DIDW centrifugal, backward curved (For FM AHU)/forward curved (For CS AHU) aerofoil blades. Fan casing shall be made of galvanized steel sheet. Fan wheels shall be made of galvanized steel. Fan shaft shall be grounded C-40 carbon steel and supported in self-aligning Plummer block

operating at less than 75% of first critical speed, grease lubricated bearings. Fan wheels and pulleys shall be individually tested and precision balanced dynamically.

Motors shall be mounted inside the AHU casing on slide rails for easy belt tensioning, and be totally enclosed, fan cooled, to be class 'F' insulation. Motors shall drive heavy duty V-belt, constant pitch, drive selected at 110% of motor horsepower.

Both fan and motors assemblies shall be mounted on a deep section aluminum alloy base frame.

Isolation shall be provided from the unit casing by combination spring and rubber anti-vibration mounts and flame retardant, waterproof neoprene impregnated flexible connection on the fan discharge.

c. Cooling Coils (DX Type):

DX Copper coils tubes with minimum 0.36 mm thick with sine wave aluminum fins firmly bonded to copper tubes assembled in zinc coated steel frame. Face and surface areas shall be such as to ensure rated capacity from each unit and such that the air velocity across the coil shall not exceed 106 meters per minute. The coil shall be pitched in the unit casing for proper drainage. The necessary valves sets including complete AHU communication kit, expansion valves etc. should be factory fitted and placed within the coil section. Single Condensate Drain Pan shall be large enough to accommodate all the split coils along with headers & valve assembly. Tube shall be mechanically expanded for minimum thermal contact resistance with fins. Fin spacing shall be 4 - 5 fins per cm.

d. Filter Section:- Min MERV-8 50mm thick prefilter section.

e. Controls:

Each AHU shall be provided with complete AHU communication Kit including DX- Type electronic expansion valve, AHU kit or any other attachment or device to make it complete and functional with the VRV outdoor unit and central controller.

3.0 Heat Recovery Unit

The Heat recovery unit shall of Double Skin with a two tier, two air stream design. The recovery wheel & casing should be from the same manufacturer.

3.1 Type:

The Heat Recovery units shall be two stream units in double skin construction, comprising of supply air section, return air section and Heat Recovery Section.

3.2 Casing:

Double skinned panel to be fabricated with anodized extruded aluminum section framework bolted together with sandwich panel having 22 gauge pre-coated outer skin and 22 gauge plain GI inner skin injected with 43 mm thick PU foam insulation having density of $38 \pm 2 \text{ kg/m}^3$. The units shall be made of extruded Aluminium hollow profile frames. The profile box size shall be of

thermal break type. The unit should be devoid of any welded construction and should be of cabinet type. All the frames should be assembled using glass fibre reinforced nylon joints/corners to make a self-supporting frame.

The Casing shall have the following parameters –

1. Deflection
2. Leakage
3. Thermal transmittance
4. Thermal Bridging.

The Inspection and access panels shall be hinged type. The hinges shall be casted, powder coated Zinc alloy. Flushed Locks and Handles shall be of glass fibre reinforced polyamide. Other panels will be screwed on to the frame with sealant and soft rubber gasket thus making the joints air tight. All screws used for panel fixing shall be covered with PVC caps.

Special hollow gaskets and seals shall be used on inspection doors and to create separation between the airstreams to ensure negligible air leakage and mixing

The entire casing shall be mounted on galvanized channel. Condensate drain pan shall be fabricated from 18 g GSS/SS construction.

3.2.1 Outdoor Version:

For Outdoor Installation, units shall have factory installed galvanized sheet metal roof. Exhaust air hood with bird screen will be provided for exhaust air & intake louvers will be provided for intake section.

3.2.2 Supply Air Section:

The supply air section shall comprise of the following:

3.3 Fan Section:

Fans shall be DIDW centrifugal, backward curved aerofoil blades. Fan casing shall be made of galvanized steel sheet. Fan wheels shall be made of galvanized steel. Fan shaft shall be grounded C-40 carbon steel and supported in self-aligning Plummer block operating at less than 75% of first critical speed, grease lubricated bearings. Fan wheels and pulleys shall be individually tested and precision balanced dynamically.

Motors shall be mounted inside the AHU casing on slide rails for easy belt tensioning, and be totally enclosed, fan cooled, to be class 'F' insulation. Motors shall drive heavy duty V-belt, constant pitch, drive selected at 110% of motor horsepower.

Both fan and motors assemblies shall be mounted on a deep section aluminum alloy base frame.

Isolation shall be provided from the unit casing by combination spring and rubber anti-vibration mounts and flame retardant, waterproof neoprene impregnated flexible connection on the fan discharge.

3.4 Motor (VFD Compatible) And Drive:

Fan motor shall be energy efficient (minimum IE-3) and suitable for 415 +/-10% volts, 50 cycles, 3 phase squirrel cage, totally enclosed fan cooled with IP 55 protections. Motor shall be designed for quiet operation.

3.5 Filter Section:

Detailed specification described in AHU sections.

3.6 Mixing Section:

The mixing section shall have built in dampers made up of aluminum aerofoil profiles with leakage Class III. The damper blades shall be controlled with plastic gear wheels and silicone gaskets shall be provided between the blades. Inspection hatch shall be provided.

3.7 Damper Section:

Damper section shall contain a built in damper of aluminum profile with leakage class III. The damper blades shall be connected with plastic gear wheels with a gasket of silicon rubber to produce tightness between the blades. The Dampers shall have provision for damper actuator mounting.

3.8 Return Air Section:

The return air section shall comprise of all above sections as in supply air section, but only filter section will be with 50mm thick prefilter section (MERV-8) instead of EAC filter and the specification for this section shall remain same as defined above.

3.9 Heat Recovery Section:

The Heat Recovery section shall include enthalpy wheels and shall have minimum recovery of 75 % of total heat, i.e both sensible and latent (each being 75 %). The wheel shall be made of pure aluminum foil coated with molecular sieve desiccant with pore diameter of 30Å. The cross contamination between the two air streams shall be nil and leakage less than 0.04%. The vertical and radial run of the wheel shall be less than 1 mm per meter of diameter. The wheels shall have noncontact labyrinth seals for effective sealing between the two air streams.

Heat Recovery Wheel specifications:

Rotor/wheel matrix shall have following Matrix:

The substrate: The substrate or wheel matrix should be made of pure aluminum foil so as to allow.

3.9.1.1.1 Quick and efficient uptake of thermal energy.

3.9.1.1.2 Sufficient mass for optimum heat transfer

3.9.1.1.3 Maximum sensible heat recovery at a relatively low rotational speed of 20 to 25rpm.

The substrate shall not be made from any material which is combustible or supports combustion like synthetic fibrous media.

The wheel shall have minimum 75% both Sensible and Latent Balanced Effectiveness as per AHRI 1060.

The wheel has to be certified as per DIN EN ISO 846 with 0% fungal and bacterial growth at 95% Relative humidity and above.

Fire rating: NFPA - 90A certification with 0% for Flame spread classification should be confirmed by manufacturer.

The Wheel shall be AHRI certified in accordance with standard 1060 and carry the AHRI certification stamp.

The product shall be produced in an ISO certified facility before execution and installation.

Necessary software selection of the wheel has to be enclosed to justify the pressure drop and efficiency calculations. The selection software shall show HRW performance in summer, monsoon & winter and approved by engineer in charge.

The Desiccant:

The desiccant should be water molecule selective and non-migratory.

The desiccant should be molecular sieve 3Å, (certified by a third party lab to have an internal pore diameter of 3Å), so as to keep the cross contamination to absolute minimum and also ensure the exclusion of contaminants from the air streams, while transferring the water vapour molecules.

The desiccant, of sufficient mass which should not be less than 5 kg per 1000 cfm of air, should be coated with non masking porous binder adhesive on the aluminum substrate so as to allow quick and easy uptake and release of water vapour. A confirmation has to be provided by manufacturer of wheel to this effect. A matrix with desiccants impregnated in non metallic substrates, such as synthetic fibre, glass fibre, etc. will not be accepted.

The rotor/wheel matrix shall have equal sensible and latent recovery.

The weight of desiccant coating and the mass of aluminum foil shall be in a ratio so as to ensure equal recovery of both sensible and latent heat over the operating range. Accordingly, a rotor matrix which has an etched or oxidised surface to make a desiccant on a metal foil and results in insufficient latent recovery and hence unequal recovery, or a rotor matrix made from desiccant integrated in a synthetic fibre matrix which result in insufficient sensible recovery, high rotation speed, and unequal recovery, will not be accepted.

Rotor:

With optimum heat and mass through matrix formed by desiccant, of sufficient mass, coated on an aluminum foil, the rotor should rotate at lower than 20 to 25 RPM, thereby also ensuring long life of belts and reduced wear and tear of seals.

The rotor shall be made of alternate flat and corrugated aluminum foil of uniform width.

The rotor honeycomb matrix foil should be so wound and adhered (full node line adhesive to be provided) so as to make a structurally very strong and rigid media which shall not get cracked, deformed etc. due to change of temperature or humidity.

The rotor having a diameter upto 2400 mm shall have spokes to reinforce the matrix. The internal threaded rod type spokes shall not be acceptable as it weakens the wheel structural strength thereby reducing the service life. From 2000 mm diameter upwards, the option of a special wing structure, to prevent

the rotors from wobbling or deforming due to the successive pressure differentials, will be available.

Sectioned wheels, with pie segments, capable of being assembled in the field, shall be available as an option, above 2600 mm in diameter.

Wheels above 3400 mm in diameter shall be segmented and provided with wing structures spokes and flanged rim. The segmented wheels shall be provided with field rotation adjustment mechanism.

The HRW hub shall be fabricated out of heavy duty steel in order to have high mechanical strength. Light duty Aluminum hub shall not be accepted.

The surface of the wheel/rotor should be highly polished (FACED) to ensure that the vertical run out does not exceed + 1 mm for every 1 meter diameter, thereby ensuring, negligible leakage, if labyrinth non contact seals are provided, and minimal drag, if contact wiper seals are provided.

The radial run out also shall not exceed + 1 mm for every 1 meter diameter, thereby minimizing the leakage/drag on the radial seals, and minimize the fluctuation in the tension of the drive belt.

The number of wraps (of alternative corrugated and flat foil) for every inch of rotor radii shall be very consistent so as to ensure uniform air flow and performance over the entire face in the air stream. Flute height and pitch will be consistent to a very tight tolerance to ensure uniform pressure drop and uniform airflows across the rotor face.

The rotor shall be a non clogging aluminum media, having a multitude of narrow aluminum foil channels, thus ensuring a laminar flow, and will allow particles upto 800 microns to pass through it.

The media shall be cleanable with compressed air, or low pressure steam or light detergent, without degrading the latent recovery.

The Cassette / casing

The recovery wheel cassette/casing shall be manufactured from tubular / sheet metal structure to provide a self-supporting rigid structure, complete with access panels, purge sector, rotor, bearings, seals, drive mechanism complete with belt. The sheet metal should be coated with a special corrosion inhibitor coating and a certificate for the same should be provided.

The rotor/wheel should have a field adjustable purge mechanism to provide definite separation of airflow minimizing the carryover of bacteria, dust and other pollutants, from the exhaust air to the supply air. It shall be possible, with proper adjustment, to limit cross contamination to less than 0.04% of that of the exhaust air concentration.

The face and radial seals shall be four (4) pass non-contact labyrinth seals / brush seals for effective sealing between the two air streams, and also for a minimum wear and tear ensuring long life of the seals.

3.10 Performance Testing:

The HRW manufacturer shall have in-house test facilities for performance testing of HRWs. If required the manufacturer shall be able to offer type testing of HRWs at their works and submit a type test report.

The manufacturer shall have test facilities for carry over testing at the same facilities where HRWs are manufactured.

SECTION -4 - VENTILATION FAN

1. General

The ventilation fans, blowers and air washer shall be complete in all respects and shall generally comply with the specifications given below:-

2. Axial Flow Fan (Standard)

- a. Tube Casing: Casing shall be constructed of heavy gauge mild steel/GSS (hot dip Galvanized with minimum 220 GSM Zinc Coating). Fan casing, motor mount shall be of bolted/welded steel construction. Motor mounting plate thickness shall be as per manufacturer standard where the motor mounting plate should take the complete load of motor and impeller assembly without vibration. An inspection view port shall be provided. Fan casing are with internal punched inlet and outlet flange to prevent air leakage. Casing shall have flanged connection on both ends for ducted applications. Support brackets for ceiling suspension shall be bolted to the casing for connection to hanger bolts. Casing shall be epoxy painted or powder coated min DFT 60micron. Casing should be long cased where impeller and motor assembly should not protrude outside the casing.
- b. Hub & Impeller: Hub and adjustable pitch blades shall be cast aluminum LM6 or cast steel construction. Blades shall be die-formed airfoil shaped for maximum efficiency and shall vary in twist and width from hub to tip to effect equal air distribution along the blade length. Fan blades mounting on the hub shall be statically and dynamically balanced the fan pitch control must be manually readjust able type. Upon installation, it require manual readjustment at site, for obtaining actual air flow values, as specified and quoted.
- c. Motor shall be energy efficient (IE2) squirrel-cage, totally-enclosed, fan cooled, standard frame, constant speed, continuous duty, single winding, suitable for $415 \pm 10\%$ volts, 50 cycles, 3 phase AC power supply, provided with class 'F' insulation. Fan shall be selected for maximum efficiency or minimum horsepower for lowest sound level (not more than 85dba @3meter from source). Motor conduit box shall be mounted on exterior of fan casing, and lead wires from the motor to the conduit box shall be protected from the air stream by enclosing in a flexible metal conduit. Motor should be licensed under IS: 12615 - Bureau of Indian Standards and shall have BIS/ISI mark and motor nameplate shall bear the license number.
- d. Mounting impeller directly on the drive shaft and secure in place with locking keyway assembly. Motor and impeller shall be removable from the inlet/outlet side of the fan as per site condition.
- e. Accessories: The following accessories shall be provided with all fans on demand:
 - Bird screen, Gravity Louver, Canvass Connection and Sound Attenuator (if required)
- f. Performance Test-Factory performance test required conforming all rated and duty parameters as per IS 3588 standard.

3. Axial Flow Fan (for Fire, Smoke & Heat Exhaust)

- a. Tube Casing: Casing shall be constructed of heavy gauge mild steel/ GSS (hot dip Galvanized with minimum 220 GSM Zinc Coating). Fan casing, motor mount shall be of bolted/welded steel construction. Motor mounting plate thickness shall be as per manufacturer standard where the motor mounting plate should take the complete load of motor and impeller assembly without vibration. An inspection view port shall be provided. Fan casing are with internal punched inlet and outlet flange to prevent air leakage. Casing shall have flanged connection on both ends for ducted applications. Support brackets for ceiling suspension shall be bolted to the casing for connection to hanger bolts. Casing shall be epoxy painted or powder coated min DFT 60micron.

Casing should be long cased where impeller and motor assembly should not protrude outside the casing.

- b. Hub & Impeller: Hub and adjustable pitch blades shall be cast aluminum LM6 or cast steel construction. Blades shall be die-formed airfoil shaped for maximum efficiency and shall vary in twist and width from hub to tip to effect equal air distribution along the blade length. Fan blades mounting on the hub shall be statically and dynamically balanced the fan pitch control must be manually readjust able type. Upon installation it require manual readjustment at site, for obtaining actual air flow values, as specified and quoted.
- c. Motor shall be energy efficient (IE2) squirrel-cage, totally enclosed non ventilated (TENV) in case motor is in air stream, standard frame, constant speed, continuous duty, single winding, suitable for $415 \pm 10\%$ volts, 50 cycles, 3 phase AC power supply, provided with class 'H' insulation. Fan shall be selected for maximum efficiency or minimum horsepower for lowest sound level (not more than 85dba @3mtr from source). Motor conduit box shall be mounted on exterior of fan casing, and lead wires from the motor to the conduit box shall be protected from the air stream by enclosing in a flexible metal conduit. Motor should be licensed under IS: 12615 - Bureau of Indian Standards and shall have BIS/ISI mark and motor nameplate shall bear the license number.
- d. The fan impeller and motor shall be direct drive with high efficiency motors in accordance with **EN12101-3:2015/ CE/UL**, foot mounted with IP55 enclosures and class H. They shall have sealed for life ball bearings. Motors shall be pre-wired to an external electrical terminal box through weatherproof flexible conduit.
- e. Fan manufacturer shall submit "certificate of conformity for 250 deg C for 2 hours operation as per EN12101-3:2015 issued by internationally recognized independent fire laboratory. The make of motor used during the testing and the make of motor supplied by the fan manufacturer at site should be same.
- f. Accessories: The following accessories shall be provided with all fans on demand:
 - Bird screen, Gravity Louver, Canvass Connection and Sound Attenuator.
- g. Performance Test-Factory performance test required conforming all rated and duty parameters as per IS 3588 standard.

4. CABINET FAN SECTION

The ventilation unit shall be of double/skin construction housing comprising of various sections such as filter section etc. fabricated of 18G galvanized steel sheet from outside. All corner supporting frames shall be of extruded Aluminum section. It shall have suitable size of inlet & outlet.

The centrifugal forward curve fan (DIDW) & motor shall be mounted on the common base of Aluminum, inside the housing including anti-vibration.

The fan outlet shall be connected to the casing with the help of fire retardant flexible canvass.

The fan shall be complete with motor, belt pulley with all accessories.

Fan motors shall be 415 | 10% volts, 50 cycles, 3 phases, squirrel – cage, totally enclosed fan cooled with IP – 54/55 protection. Motor shall be especially designed for quiet operation and motor speed shall not exceed 1440 RPM. Drive to fan shall be provided.

Small capacity Fan, motors shall be 220 | 10% volts, 1 phase,

Each unit shall be provided with a factory assembled filter section with 50 mm thick washable synthetic type air filters having efficiency 90% down to 10 micron with extruded aluminum frame.

The opening for access doors and gaps between sections shall be provided with the neoprene rubber T gasket fixed in the extruded sections.

In case of exhaust, the fan outlet shall have Louvers with bird screen.

The cabinet for fan should be as per AMCA Standard.

5. Blower drive assembly

Drive assembly for each blower shall consist of blower pulley, motor pulley, a set of 'V' belts, belt guards, and belt tension adjusting device.

Pulleys shall be selected to provide the required speed. They shall be multi-groove type, with section and grooves selected to transmit 33% more load than the required power and shall be statically balanced.

The belt guards shall be of M.S. Sheet with angle iron reinforcements and expanded metal screen.

Motors and starters

The motor for each blower shall be squirrel cage high efficiency (efficiency -1) induction type and conform to specifications as given under section on control panel, motors and switchgear. The motor H.P. Shall be at least 20% more than the limit load of fan and of minimum rating as given under 'Schedule of Equipments'. The Motors shall be as per IS-325-1996 (Revised to date) with F-Class of Insulation.

6 Accessories

All necessary accessories shall be provided for proper operation and shall also include **(As part of**

Unit Price).

- Dunlop cushy foot vibration isolators for the blowers.
- Double canvass connections at the outlet of each fan.
- Nuts, bolts, shims etc. as required for the grouting of the equipment.
- Slide rails for mounting the motor and belt adjustments.
- Bird Screens in the Inlet.
- Louvers for Fresh Air and exhaust openings.
- Detachable and washable fresh air filters at the inlet.

7. INLINE FANS

7.1 Circular Inline Fans

Circular inline Centrifugal duct fan shall be a straight through radial fan. It should be compact. Motor shall be energy efficient.

The fan shall cope with high pressure and long duct runs, whilst still operating at an acceptable sound level.

The circular inline fans should be moisture resistant and should be approved for installing in humid or damp environments. The fans should be rated IP-44 when installed in a duct system. The casing should be manufactured from pre galvanized steel. Automatic thermo – contact shall open up if the temperature within the motor windings becomes excessive. Fan should be equipped with external rotor single phase asynchronous motor.

7.2 Cabinet Inline Fans (Above 1000 CFM capacity)

The Cabinet of the cabinet inline fans should be manufactured in galvanized steel & shall be with insulated housing for lower noise levels or shall be double skin. The Mounting brackets and access panel should be fitted as per manufacturer standard. The impeller of the fans should be with high efficiency forward/backward curved DIDW/SISW type. The scroll of the fans should be fabricated with galvanized sheet steel. The motor of the fans shall be Squirrel cage induction type with bearings sealed for life. The motor shall be energy efficient suitable for single phase electrical supply. The cabinet of the fan shall be of double skin construction with acoustically lined with 25 mm glass wool with perforated GI sheet from inside. Fan noise level shall not be more than 55dBA at 3 meters distance in hemispherical reverberant room conditions.

8. PERFORMANCE DATA

All fans shall be selected for the lowest operating noise level. Capacity ratings, power consumption, with operating points clearly indicated, shall be submitted and verified at the time of testing and commissioning of the installation.

9. TESTING

Capacity of all fans shall be measured by an anemometer. Measured air flow capacities shall conform to the specified capacities and quoted ratings. Power consumption shall be computed from measurements of incoming voltage and input current.

10. INSTALLATION

- i) The EPC Contactor shall supply all required bolts, base frame (wherever required), vibration isolators any other accessories and shall assure that the components are placed securely in proper position.
- ii) Vibration isolators shall be provided with an efficiency of not less than 80%.

11. PROPELLER FAN

Propeller fan shall be direct-driven, multiple blade type, mounted on a steel mounting plate with orifice ring.

- a. Mounting Plate shall be of steel construction, square with streamlined venturi inlet (reversed for supply applications) coated with baked enamel paint or with min 220gsm zinc coating. Mounting plate shall be of standard size depending upon the fan size as per OEM. Orifice ring shall be correctly formed by spinning or stamping to provide easy passage of air without turbulence and to direct the air stream.
- b. Fan Blades shall be constructed of aluminium or steel. Fan hub shall be of heavy welded steel construction with blades bolted to the hub. Fan blades and hub assembly shall be statically and dynamically balanced at the manufacturer's works.
- c. Shaft shall be of steel, accurately ground and shall be of ample size for the load transmitted and shall not pass through first critical speed thru the full range of specified fan speeds.
- d. Motor shall be energy efficient standard (easily replaceable) permanent split capacitor or shaded pole for small sizes, totally enclosed with prelubricated sleeve or ball bearings, designed for quiet operation as per approved design for $220 \pm 10\%$ volts, 50 cycles single phase power supply/ $415 \pm 10\%$ volts, 50 cycles three phase power supply.
- e. Accessories : The following accessories shall be provided with propeller fans:
 - i. Wire guard on inlet side and bird-screen at the outlet.
 - ii. Fixed or gravity louvers built into a steel frame at the outlet.

SECTION – 5

DUCT WORK AND OUTLET

1. GENERAL

This section comprises of supply, fabrication, installation and testing of all sheet metal ducts and supply, installation, testing and balancing of grilles, registers and diffusers, in accordance with these specifications.

2. MATERIAL

2.1. Ducts

- i) All ducts shall be fabricated from Galvanised Sheet Steel (GSS) conforming to IS: 277. The steel sheets shall be hot dip galvanized with MAT finish with coating of minimum 120 grams per square meter (GSM) of Zinc, GI sheets shall be lead free. Eco friendly and RoHS compliant.
- ii) The thickness of sheets for fabrication of rectangular ductwork shall be as under. The thickness required corresponding to the longest side of the rectangular section shall be applicable for all the four sides of the duct work.

Longest side (mm)	Minimum sheet thickness
	For GSS
750 mm and below	0.63
751 mm to 1500 mm	0.80
1501 mm to 2250 mm	1.00
2251 mm & above	1.25

- (iii) Thickness of sheet for Round Ducts

Diameter of duct, mm	Thickness of Sheet, mm
	GI sheets
150 to 500	0.63
501 to 750	0.80
751 to 1000	0.80
1001 to 1250	1.00
1251 and above	1.25

- iv) All sheet metal connections, partitions and plenums required for flow of air through the filters, fans etc. shall be at least 1.25 mm thick galvanised steel sheets, in case of G.I. sheet ducting or 1.8 mm thick aluminium sheet, in case of aluminium sheet ducting and shall be stiffened with 25 mm x 25 mm x 3 mm angle iron braces.
- v) Circular ducts, where provided shall be of thickness as specified in IS: 655 as amended upto date.
- vi) Aluminium ducting shall normally be used for clean room applications, hospitals works and wherever high cleanliness standards are functional requirements.

2.2 Associated Items

- Supply/return air outlets, F.A. grilles and accessories shall be constructed from extruded aluminium sections.
- Flanges for matching duct sections, stiffening angles (braces) and supporting angles shall be of rolled steel sections, and shall be of the following sizes.

Application	Duct Width	Angle size
Flanges	Upto 1000 mm	35 mm x 35 mm x 3 mm
-do-	1001 mm to 2250 mm	40 mm x 40 mm x 3 mm
-do-	More than 2250 mm	50 mm x 50 mm x 3 mm
Bracings	Upto 1000 mm	25 mm x 25 mm x 3 mm
-do-	More than 1000 mm	40 mm x 40 mm x 3 mm
Support angles	Upto 1000 mm	40 mm x 40 mm x 3 mm
-do-	1001 mm to 2250 mm	40 mm x 40 mm x 3 mm
-do-	More than 2250 mm	Size and type of RS section shall be decided in individual cases

2.3 CONSTRUCTION

2.3.1 Ducts

- 2.3.1.1 Ducts shall be fabricated at site or factory fabricated and shall be generally as per IS: 655 "Specifications for metal air ducts", unless otherwise deviated in these General Specifications.
- 2.3.1.2 The interior surfaces of the ducting shall be smooth.
- 2.3.1.3 All the ducts upto 600 mm longest side shall be cross broken between flanges by a single continuous breaking. Ducts of size 600 mm and above shall be cross broken by single continuous breaking between flanges and bracings. Alternatively, beading at 300 mm centres for ducts upto 600 mm longest side, and 300 mm centres for ducts above 600 mm size shall be provided for stiffening.
- 2.3.1.4 As far as possible, long radius elbows and gradual changes in shape shall be used to maintain uniform velocity accompanied by decreased turbulence, lower resistance and minimum noise. The ratio of the size of the duct to the radius of the elbow shall be normally not less than 1:1.5.
- 2.3.1.5 Flanged joints shall be used at intervals not exceeding 2500 mm. Flanges shall be welded at corners first and then riveted to the duct.
- 2.3.1.6 Stiffening angles shall be fixed to the sides of the ducts by riveting at 1.25 meters from joints for ducts of size 600 mm to 1500 mm, and 0.6 m from joints for ducts of size larger than 1500 mm. Bracings for ducts larger than 1500 mm can alternatively be by diagonal angles.
- 2.3.1.7 Plenums for filters shall be complete with suitable access door of size 450 mm x 450 mm.
- 2.3.1.8 All factory fabricated duct shall be supplied in L sections, the length of any piece shall not be more than 1800 mm for duct with longest side of cross section as 600 mm and above and 3000 mm for rest.

2.3.2 Air Outlet and Inlets (Supply and Return)

- 2.3.2.1 All air outlets and intakes shall be made of extruded aluminium sections & shall present a neat appearance and shall be rigid with mechanical joints.

- 2.3.2.2 Square and rectangular wall outlets shall have a flanged frame with the outside edges returned or curved 5 to 7 mm and fitted with a suitable flexible gasket between the concealed face of the flanges and the finished wall face. The core of supply air register shall have adjustable front louvers parallel to the longer side to give up to 22.5 degrees vertical deflection and adjustable back louvers parallel to the shorter side to achieve a horizontal spread air pattern to at least 45 degrees. Return air grilles shall have only front louvers. The outer framework of the grilles shall be made of not less than 1.6 mm thick aluminium sheet. The louvers shall be of aerofoil design of extruded aluminium section with minimum thickness of 0.8mm at front and shall be made of 0.8mm thick aluminium sheet. Louvers may be spaced 18 mm apart.
- 2.3.2.3 Square and rectangular ceiling outlets/intakes shall have a flange flush with the ceiling into which it is fitted or shall be of anti smudge type. The outlets shall comprise an outer shell with duct collar and removable diffusing assembly. These shall be suitable for discharge in one or more directions as required. The outer shell shall not be less than 1.6 mm thick extruded section aluminium sheet. The diffuser assembly shall not be less than 0.80 mm thick extruded aluminium section.
- 2.3.2.4 Circular ceiling outlets/intakes shall have either flush or anti smudge outer cone as specified in the tender specifications. Flush outer cones shall have the lower edge of the cone not more than 5 mm below the underside of the finished ceiling into which it is fitted. Anti smudge cones shall have the outer cone profile designed to reduce dirt deposit on the ceiling adjacent to the air outlet. The metal sheet used for construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.
- 2.3.2.5 Linear diffusers shall have a flanged frame with the outside edges returned 3.5 mm and shall have one to four slots as required. The air quantity through each slot shall be adjustable. The metal sheet used for the construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.
- 2.3.2.6 Grilles and diffusers constructed of extruded aluminium sections shall have grille bars set straight, or deflected as required. These shall be assembled by mechanical interlocking of components to prevent distortion. These grilles and diffusers shall have a rear set of adjustable blades, perpendicular to the face blades for deflection purposes.
- 2.3.2.7 All supply air outlets shall be fitted with a volume control device, made of extruded aluminium gate section. The blades of the device shall be mill finish/ block shade pivoted on nylon brushes to avoid rusting & rattling noise, which shall be located immediately behind the outlet and shall be fully adjustable from within the occupied space without removing any access panel. The volume control device for circular outlets shall be opposed blade radial /shutter type dampers, or two or more butterfly dampers in conjunction with equalizing grid. Opposed blade dampers shall be used for square and rectangular ceiling/wall outlets and intakes.
- 2.3.2.8 All the products supplied by contractor should supplement in performance by selection curves of product ratings from the manufacturer.

3. FRESH AIR INTAKES

- Fresh air intake grills shall be made of extruded aluminium sections.
- A flanged frame using RS sections shall be provided on front face to conceal the gap

between the louvers and the adjoining wall face. Corners of frame shall be welded. The frame shall be made structurally rigid.

- Louvers made from extruded aluminum section shall be in modular panel form for ease of handling. These shall be free from waves and buckles. Vertical blades shall be truly vertical and horizontal blades shall be truly horizontal. Butt joints in blades shall not be accepted.
- Additional intermediate equally spaced supports and stiffeners shall be provided to prevent sagging/ vibrating of the louvers, at not more than 750mm centres where the louver's length is longer than 750mm.
- A bird wire screen made of 12 mm mesh in 1.6 mm steel wire held in angle or channel frame shall be fixed to the rear face of the louver frame by screens.

4. FIRE DAMPERS

- i) Fire dampers shall be provided in all the supply air ducts and return air ducts (where provided), return air passage in the air-handling unit room and at all floor crossings. Access door will be provided in the duct before each set of fire dampers.
- ii) Fire dampers shall be multi blade louvers type. The blade should remain in the airstream in open position & shall allow maximum free area to reduce pressure drop & noise in the air passage. The blades and frame shall be constructed with minimum 1.6mm thick galvanised sheet & shall be factory fitted in a sleeve made out of 1.6mm galvanised sheet of minimum 400mm long. It shall be complete with locking device, motorised actuator & control panel.
- iii) Fire dampers shall be motorised smoke & fire dampers type. It shall be supplied with spring loaded IS/ UL stamped fusible link to close fire damper in the event of rise in duct temperature. Fire damper shall also close on receipt of fire alarm signal to cut off air supply instantaneously. An electric limit switch shall also be operated by the closing of fire damper, which in turn shall switch off power supply to AHU blower motor as well as strip heaters.
- iv) Fire dampers shall be I S / CBRI/UL tested & certified for 90 minutes rating against collapse & flame penetration as per IS/UL 555-1995.(Under writers laboratories)
- v) Fire dampers shall be compatible with the fire detection system of building & shall be capable of operating automatically through an electric motor on receiving signal from fire alarm panel.
- vi) Necessary wiring from fire alarm panel up to AHU electric panel shall be provided by the department & further from AHU electric panel to fire damper shall be provided by air conditioning contractor.

5. Air Outlet and Inlets (Supply and Return)-GRILL/ DIFFUSER

All air outlets and intakes shall be made of extruded aluminium sections & shall present a neat appearance and shall be rigid with mechanical joints. Square and rectangular wall outlets shall have a flanged frame with the outside edges returned or curved 5 to 7 mm and fitted with a suitable flexible gasket between the concealed face of the flanges and the finished wall face. The core of supply air register shall have adjustable front louvers parallel to the longer side to give upto 22.5 degrees vertical deflection and adjustable back louvers parallel to the shorter side to achieve a horizontal spread air pattern to at least 45 degrees.

Return air grilles shall have only front louvers. The outer framework of the grilles shall be made of not less than 1.6 mm thick aluminium sheet. The louvers shall be of aerofoil design of

extruded aluminium section with minimum thickness of 0.8mm at front and shall be made of 0.8mm thick aluminium sheet. Louvers may be spaced 18mm apart.

Square and rectangular ceiling outlets/intakes shall have a flange flush with the ceiling into which it is fitted or shall be of anti smudge type. The outlets shall comprise an outer shell with duct collar and removable diffusing assembly. These shall be suitable for discharge in one or more directions as required. The outer shell shall not be less than 1.6 mm thick extruded section aluminium sheet. The diffuser assembly shall not be less than 0.80 mm thick extruded aluminium section.

Circular ceiling outlets/intakes shall have either flush or anti smudge outer cone as specified in the tender specifications. Flush outer cones shall have the lower edge of the cone not more than 5 mm below the underside of the finished ceiling into which it is fitted. Anti smudge cones shall have the outer cone profile designed to reduce dirt deposit on the ceiling adjacent to the air outlet. The metal sheet used for construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.

Linear diffusers shall have a flanged frame with the outside edges returned 3.5 mm and shall have one to four slots as required. The air quantity through each slot shall be adjustable. The metal sheet used for the construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.

Grilles and diffusers constructed of extruded aluminium sections shall have grille bars set straight, or deflected as required. These shall be assembled by mechanical interlocking of components to prevent distortion. These grilles and diffusers shall have a rear set of adjustable blades, perpendicular to the face blades for deflection purposes.

All supply air outlets shall be fitted with a volume control device, made of extruded aluminium gate section. The blades of the device shall be mill finish/ block shade pivoted on nylon brushes to avoid rusting & rattling noise, which shall be located immediately behind the outlet and shall be fully adjustable from within the occupied space without removing any access panel. The volume control device for circular outlets shall be opposed blade radial /shutter type dampers, or two or more butterfly dampers in conjunction with equalizing grid. Opposed blade dampers shall be used for square and rectangular ceiling/wall outlets and intakes.

All the products supplied by contractor should supplement in performance by selection curves of product ratings from the manufacturer.

Laminar supply air diffusers shall be made of 2mm thick powder coated aluminium sheet duly insulated with 5mm thick closed cell polyethylene foam insulation having factory laminated aluminium foil and joints covered with self adhesive aluminium tape and having holes 2/3 mm dia including frame work.

6. JET DIFFUSER/NOZZLE (Eye Ball type)

Jet diffusers shall be designed to handle high air flow rates and provide relatively long throws which makes them particularly suitable for conditioning of large spaces such as halls, auditoria, terminal buildings. The diffusers shall be versatile in application by virtue of the novel "reversible and rotatable" core design which allows the air jet to be adjusted for both pattern and deflection. Units may be mounted individually or in banks in bulkhead arrangements or directly into stub ducts.

Units shall be constructed from aluminium spinnings supported on a studding and spacer assembly. The core may be rotated through 180 degrees to expose either a straight or diffused core assembly. In addition, the diffuser spigot may be rotated within the mounting plate (if supplied) to enable a full 360-degree adjustment, by releasing the tension bolts.

It should have following Features:

- Simple and effective air diffusion for large spaces.
- Long throw characteristics.
- High air handling capacity.
- Easy to install and adjust.
- Reversible core to produce long throw jet or short throw diffuser patterns.
- Core rotatable through 360 degrees for "eyeball" jet direction.
- Jet may be deflected off axis by up to 15 degrees.

The **JET NOZZLE** shall be aerodynamically designed powder coated aluminium construction double skin eye ball type with double flap butterfly damper, mounted on steel face plate suitable for side wall installation. Nozzle shall be deep drawn in one single pcs with discharge nozzle and face cover ring crimped on nozzle base to avoid the misplacing of ring problem. The Nozzle shall be housed in a specially designed housing with a mounting flange with holes for fastening the nozzle and aesthetic trim ring with snap locking arrangement to cover fasteners and give blemish free appearance. The color shade shall be as desired by Client/Architect/AAI.

Type- jet nozzles eye ball

Draft length- 30mtr (as per requirement) Deflection – 0 to 40 degree

Type of finish – powder coated Noise level – 30 NC

Housing – aluminum finish with snap locking face ring

7. OUTSIDE AIR LOUVERS

Exhaust/Fresh air louvers of 80 mm thick high performance (55% free area) drainable fixed louver type powder coated Aluminium frame and blades. Mullions to be sliding interlock type with integral internal drain. Jamb and mullion drains to be open on front face in order to direct water away from inside of louver. Blades to be one piece extrusions with gutters design to catch and direct water to jamb and mullion drains. Fasteners to be aluminium. Louvers to have framed 13 mm mesh removable mill finish aluminium bird screens.

8. DUCTING SUPPORTS:

- a) All ductwork shall be independently supported from building construction. All horizontal ducts shall be rigidly and securely supported, in an approved manner, with hangers formed of galvanized steel wire ropes and galvanized steel angle/channel or a pair of brackets, connected by galvanized steel wire hangers under ducts, rigid supports may be provided at certain interval if need be. The spacing between supports should be not

greater than 2.4 meter. All vertical ductwork shall be supported by structural/strut members on each floor slab. Duct supports may be through galvanized steel insert plates or Toggle end wire fixing left in slab at the time of slab casting. Galvanized steel cleat with a hole for passing the wire rope hanger shall be welded to the plates. Trapeze hanger formed of galvanized steel wire rope using hang fast shall be hung through these cleats. Wherever use of metal insert plates is not feasible, duct support shall be through dash/anchor fastener driven into the concrete slab by electrically operated gun. Wire rope supports shall hang through the cleats or wire rope threaded studs can be screwed into the anchor fasteners. In case of PEB structure Loop and Catenary system can be used based on the site conditions as per approved suspension system drawings.

- b) All horizontal ducts shall be adequately secured and supported. In an approved manner, with trapeze Hangers formed of galvanized steel wire rope in a cradle support method under ducts at no greater than 3000mm centre, for 3001mm-above appropriate size angle along with neoprene pad in between the duct & MS angle should be used with prior approval. All vertical duct work shall be supported by structural members on each floor slab. Duct support shall be through dash/anchor fastener driven into the concrete slab by electrically operated gun. Hanger wires shall then hang around the ducting. Rigid supports shall be used in conjunction with wire rope hangers to assist with alignment of services where recommended for by the manufacturer. Rigid support must also be used in conjunction with wire rope hangers with duct work at each change of direction or connection. Support ducting in accordance with Schedule I at the end of this Section. Any other Wire hanging arrangement solution can be used based on manufacturer's recommendation on site conditions after prior approval. In cases of Spiral ducting the wire can be wrapped directly around the ducting without the need for a spiral ducting clamp for sizes above 1100 a cradle support should be provided refer to manufacturer's recommendations.
- c) Ducting over furred ceiling shall be supported from the slab above or from beams after obtaining approval of Construction manager/consultant. In no case shall any duct be supported from false ceiling Hangers or be permitted to rest on false ceiling. All metal work in dead or furred down spaces shall be erected in time to occasion no delay to work in the building. All supports of pipe shall be taken from structural slab/wall by means of fastener.

Catenary Supports:

Refer to manufacturer's recommendations on Catenary supports with C-clip, special care should be taken with tensioning of the wire and angles at which the installation of services are made.

d) Duct Installations (As per NBC 2016 (Part – 8.3) Clause-12.1.7.1)

Comply with manufacturer's load ratings and recommended installation procedures.

Schedule I: Duct Hanger Schedule

For ducts with external SP upto 250 Pa		
Maximum Duct Size (mm)	Gauge	Hanger No.
1 - 750	26	1/2

751-1000	26	2
1001-1200	24	2/3
1201 - 1500	24	3
1501 - 1800	22	3/4
1801-2100	20	3/4
2101-2700	18	4

For ducts with external SP upto 500 Pa

Maximum Duct Size (mm)	Gauge	Hanger No.
1-600 mm	26	1/2
601-750 mm	26	2
751-1000 mm	24	2/3
1001-1200 mm	22	3/4
1201-1300 mm	20	3/4
1301-1500 mm	18	4
1501-1800 mm	18	4
1801-2100 mm	18	4
2101-2250 mm	18	4/5
2251-2400 mm	18	4/5
2401-2700 mm	18	4/5

Notes: All supports are considered at 2400 mm interval in above table and may vary as per the design but should not be greater than 2400 mm.

9. FLEXIBLE DUCTING

Insulated flexible duct should be IS/UL 181 CLASS I AIR DUCT LISTED AND LABELLED WITH NFPA 90A & 90B AND SEAL OF AIR DIFFUSION COUNCIL with double lamination of tough polyester which encapsulates steel helix wire forms the air tight inner core, double layer core wrapped in a multiple thickness of fiberglass wool with R Value 4.2, Green guard certification of fiberglass wool must. Reinforced and sheathed in a rugged and durable tridirectionally reinforced metalized polyester jacket. Flexible duct connections should be made as per IS/UL181 listing procedure with proper flexible right forming brace connection allowing right connections for flexible duct into energy efficient. and Strapping the flexible duct connections with flexible duct strap ties.

10. BALANCING

Air systems shall be balanced in a manner to minimize throttling losses. The entire air distribution system shall be balanced with the help of an anemometer. The measured air quantities at fan discharge and at the various outlets shall be within ± 5 percent of those specified/quoted. For fans greater than 0.75 KW (1.0 HP), fans must then be adjusted to meet design flow conditions. Branch duct adjustments shall be permanently marked after the air balancing is completed so that these can be restored to their correct position if disturbed at any time.

11. Flexible Canvass Connections (For Fans, AHUs etc.)

Flexible canvass connections to isolate vibrations produced by Air handling units, Axial Flow fans and other air moving equipments when these are to be connected to air ducts where ever specified. Canvass connections shall be of the same cross-sectional area as the mating fan inlet/outlet or duct section.

Flexible canvass connection shall be provided with 45 mm galvanized metal strip of minimum 0.40 mm thickness both side with minimum 0.80 mm this fabric for easy connection to duct or equipments. The fabric shall be fire resistant, fiberglass weave with silicone rubber coating on both side. Flexible connector fabric shall withstand severe effects of strong sunlight, weathering and ageing. Flexible duct connector fabric shall be air tight, waterproof, virtually chemically inert and shall not emit toxic fumes when exposed to open flame. The flexible canvass connection shall be suitable to withstand the positive and negative fan pressures at point of installation to which they will be subjected to and shall not allow perceptible leakage and shall be completely free from stress and shall not be required to accept any weight.

Duct work connections to the fan inlets / outlets shall be concentricity aligned so that the flexible connections are not subjected to any strain

Flexible Duct connector shall be tested and certified in accordance with BS 476 part 7 for class 1 and NFPA -701 from NABL accredited lab

SECTION – 6 THERMAL INSULATION

1.0 SCOPE

The scope of this section comprises the supply and application of insulation conforming to these specifications. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber Foam Insulation.

2.0 MATERIAL

Thermal insulation material for Duct & Pipe insulation shall be Closed Cell Nitrile Rubber insulation. Thermal conductivity as per **DIN EN 12667 / EN ISO8497** of the insulation material shall not exceed 0.038 W/m²K or 0.212 BTU / (Hr-ft²-°F/inch) at an average temperature of 30°C. Density of the nitrile rubber shall be 40-60 Kg/m³, The product shall have temperature range of -40°C to +105°C as per EN 14706, EN 14707 and EN 14304. The insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test. The material also pass UL 94 Test for Vertical Burning and Horizontal Burning and FM Approved. Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10⁻¹³ Kg/m.s.Pa i.e. $\mu \geq 7000$: Water vapour diffusion resistance) as per **EN 12086 & EN13469**.

For Non-Exposed Ducts, the Nitrile Rubber insulation shall be with factory laminated embossed aluminium foil of minimum 60 micron thickness having weight of 193 gsm as per EN ISO 2286-2.

For Exposed Ducts the Nitrile Rubber insulation shall be with factory laminated with 3-layered non-metallic composite cladding of minimum 230 micron thickness of metallic appearance with aluminum as an intermediate layer and tested for UV resistance as per EN ISO 4892-2 Method A.

For refrigerant & drain piping:- **Insulation shall be in tube form for pipe nb dia upto 80mm and above 80mm sheets shall be used for piping.**

Ducting insulation thickness shall be as per table below.

Ducting position	Thk. Of Insulation
SA duct in RA path	19 mm
Ducted return air system	SA duct: 25 mm RA duct: 19 mm
Both SA & RA exposed	Both 25 mm

3.0 DUCT INSULATION

External thermal insulation shall be provided as follows :

The thickness of insulation material shall be as shown on drawings or identified in the schedule of quantity. Following procedure shall be adhered to:

Comply

Noted



Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90° cut in thickness of sheet. Hackshaw or blades are not acceptable tools for cutting the insulation.

Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Self Adhesive aluminum tapes / silver tapes as per the insulation material finish as per manufacturer recommendations. The insulation installers shall be certified by manufacture.

Direct contact between Duct and hanger shall be avoided. Hangers shall pass outside the Duct Support. Insulation Manufacturer shall supply duct supports having PET foam as load taking member embedded in flexible elastomeric foam nitrile rubber sheet with outer metal face having aluminium metal) so that the insulation material is joint with insulation material on both side of pipe support for Nitrile and the weight of pipe is transferred to the PET saddle.

4.0 PIPING INSULATION

All refrigerant, and condensate drain piping shall be insulated in the manner specified herein. Before applying insulation, all pipe shall be brushed and cleaned. All MS pipes shall be provided with a coat of zinc chromate primer. Thermal insulation shall be applied as follows or as specified in drawings or schedule of quantity:

Piping Insulation thickness shall be as defined in BOQ.

Insulating material in tube form (minimum upto 80 dia pipes) shall be sleeved on the pipes. On piping, slit opened tube from insulating material shall be placed over the pipe and adhesive shall be applied as suggested by the manufacturer. Adhesive must be allowed to tack dry and then press surface firmly together starting from butt end and working towards centre. Wherever flat sheets shall be used it shall be with self-adhesive and cut out in correct dimension using correct tools. Scissors or Hacksaw-blade shall not be allowed. All longitudinal and transverse joints shall be sealed by providing 50 mm wide self adhesive silver tapes as per manufacturer recommendations. The adhesive shall be strictly as recommended by the manufacturer. The insulation shall be continuous over the entire run of piping, fittings and valves. All valves, fittings, joints, strainers etc. in chilled water piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced.

Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside the saddle. Insulation Manufacturer shall supply PUF saddles with pre-laminated insulation sheet of both side (PUF saddle sandwich between nitrile insulation material on both side with outer metal shell made of aluminium) so that the insulation material is joint with insulation material on both side of pipe support for Nitrile and the weight of pipe is transferred to the PUF saddle in the center.

The non-mettalic cladding material is to be applied with 30 – 50mm overlap to be fixed with plastic rivets and finished with a silver self-adhesive tape. All bends, T-

Sections and end caps shall be clad with thermoformed fittings for pipes of upto 80mm dia. For higher dia pipes, the cladding sheet is to be cut in required shape and applied on insulation material.

Manufacturer's installation manual shall be submitted and followed for full compliance. All insulation work shall be carried out by skilled workmen specially trained and certified by manufacturer in this kind of work. All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding. Direction of fluid shall also be marked. Un-insulated MS pipes shall be painted throughout and direction of fluid marked. All painting shall be as per relevant BIS codes.

5.0 MEASUREMENT OF INSULATION

Unless otherwise specified measurement for duct and pipe insulation for the project shall be on the basis of centre line measurements described herewith

- a. Pipe Insulation shall be measured in units of length along the centre line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the centre line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever.
- b. Duct Insulation and Acoustic Lining shall be measured on the basis of surface area along the centre line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising centre line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the centre-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting.

Note: The Insulation manufacturer has to send their engineers at site and train the insulation contractors ensuring proper application and workmanship of the insulation material.

BOARD **ACOUSTIC LINING: RESIN BONDED GLASS WOOL INSULATION**

1.1 Material Specification

Material for acoustic lining shall be Resin bonded glass wool insulation board. The thermal conductivity shall not exceed 0.490 W/m-K at 25°C and density shall not be less than 80 Kg./CuM and shall be factory laminated with one side aluminium foil & other side Black Glass Cloth. Thickness of acoustic lining shall be 25 mm for duct lining and for AHU room lining.

Acoustical lining of duct shall be done for the initial run of 4.0 meters duct.

DATA SHEET: ACOUSTIC BOARD

Parameter	Technical Specification (Acoustic insulation)
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Material	Resin bonded glass wool
Standard/Codes	IS 8183: 93 /IS 3144: 90
Nominal bulk density	80 Kg/m ³
Thickness	25 mm
R-value in sq.mK/W (at 25 mm Thickness)	R-0.510
Facings	Factory lamination with one side FSK (Aluminum foil) & other side BGC (Black Glass Cloth)
Thermal conductivity	0.490 W/m-K at 25°C
Parameter	Conformance
Fire Properties	
Non combustibility	As per BS 476 part 4
Ignitibility	As per BS 476 part 5
Surface Spread of	Class 1 as per BS 476 part 7
Fire propagation	Class 'O', [index I<12, sub-index i1<6] as per BS 476 part 6 and 7.
Toxicity	As per BS 6853
Moisture content	As per IS 3144/IS 8183
Shot content	As per IS 3144/IS 8183
Sulphur /Chloride content	As per IS 3144/IS 8183
Mould growth	As per IS 3144/IS 8183. Should not support mould growth.
Chemical	PH value should be 7.

1.2 Method Of Application On Ducts And In AHU Rooms

- 1 The surface shall be cleaned and put rubber based adhesive on duct/wall and Acoustic Board (on Aluminium facing).
- 2 After 10-15 minutes of the coat of adhesive the Acoustic board has to be placed on the duct/wall and it has to be fixed with the help of Anchor/Screw and Washer.
1. The Acoustic board should be placed in such a way that black glass cloth is visible from inside the duct/AHU room.

SECTION – 7-TESTS AT SITE & FACTORY

1. GENERAL

The contractor must perform all inspection and tests of the system as a whole and of components individually as required, under the supervision of the purchasers/ Engineer-in-charge in accordance with the provisions of the applicable ASHRAE / SMACNA / ARI standards or approved equal and as per load requirements.

2. FOR ASSOCIATES WORKS AT SITE

- 2.1 All electrical items will be subjected to inspection at any stage during manufacturing activity. Routine electrical test as per relevant codes. Inspection of manufacturer's test certification.
- 2.2 Inspection of raw materials to be used for fabrication and assembly and inspection of manufacturer's certificates.
- 2.3 Checking of electrical circuits (power & controls) and checking functioning of controls and other circuits of ventilation system.
- 2.4 Checking of assemblies for electrical control panel, instruments panels, local panels (dimensional and functional) annunciator panels etc.
- 2.5 Inspection of complete electrical installation at site.
- 2.6 Performance testing of complete Ventilation system as per specifications.

- 3. The above inspection procedure is given for general guidance and information of vendors and inspection of purchaser is strictly not limited to these and inspection engineer of purchaser will have full right to have detailed inspection at any stage right from placement of order to completion of project as desired by inspection engineer, co-ordination of inspection agency of purchaser with his factory/sub-contractor's factory/ erection site will be the sole responsibility of successful contractor after placement of order for complete air conditioning plant covered under these technical specifications.

4. DUCT WORK

- 4.1 All branches and outlets shall be tested for air quantity, and the total of the air quantities shall be within plus five percent (5%) of fan capacity.
- 4.2 Fire dampers, volume dampers and splitter dampers shall be tested for proper operation.

5. BALANCING AND ADJUSTMENT

All air handling ventilation equipment, duct work and outlets shall be adjusted and balanced to deliver the specified air quantities indicated, at each inlet and outlet, on the drawings. If these air quantities cannot be delivered without exceeding the speed range available horse power, the department shall be notified before proceeding with the balancing of air distribution system.

6. ELECTRICAL EQUIPMENT

- 6.1 All electrical equipment shall be cleaned and adjusted on site before application of power.
- 6.2 The following tests shall be carried out :
 - 6.2.1 Wire and cable continuity tests.
- 6.3 Insulation resistance tests, phase to phase and phase to earth, on all circuits and equipment, using a 500 volt meggar. The meggar reading shall be not less than one meg. ohm.
- 6.4 Earth resistance between conduit system and earth must not exceed half (1/2) Ohm.
- 6.5 Phasing out and phase rotation tests.
- 6.1 Operating tests on all protective relays to prove their correct operation before engineering the main equipment.
- 6.7 Operating tests on all starters, circuit breakers, etc.

7. PERFORMANCE TESTS

- 7.1 The installation as a whole shall be balanced and tested upon completion, and all relevant information, including the following shall be submitted to the purchasers.
 - 7.1.1 Air volume passing through each unit, duct, grilles and apertures.
 - 7.1.2 Differential pressure readings across each fan.
 - 7.1.3 Electrical current readings, in amperes of full and average load running, and starting, together with name plate current of each electrical motor.
 - 7.1.4 Continuous recording over a specified period, of ambient wet and dry bulb temperatures under varying degrees of internal heat loads and use and occupation, in each zone of each part of the building.
 - 7.1.5 Daily records should be maintained of hourly readings, taken under varying degrees of internal heat load and use and occupation, of wet and dry bulb temperatures. The current and voltage drawn by each machine.
 - 7.1.6 Any other readings shall be taken which may subsequently be specified by the client.

8. MISCELLANEOUS

- 8.1 The above tests are mentioned herein for general guidance and information only but not by way of limitation to the provisions of conditions of contract and specification.
- 8.2 The date of commencement of all tests listed above shall be subject to the approval of the purchaser, and in accordance with the requirements of this specification.
- 8.3 The contractor shall supply the skilled staff and all necessary instruments and carry out any test of any kind on a piece of equipment, apparatus, part of system

or on a complete system if the purchaser requests such a test for determining specified or guaranteed data as given in the specification or on the drawings.

- 8.4 Any damage resulting from the tests shall be repaired and/or damaged material replaced, to the entire satisfaction of the department.
- 8.5 In the event of any repair or any adjustment having to be made, other than normal running adjustment, the tests shall be void and shall be recommended after the adjustment or repairs have been completed.
- 8.6 The contractor must inform the department when such tests are to be made, giving sufficient notice, in order that the purchaser or his nominated representative may be present.
- 8.7 Complete records of all tests must be kept and 3 copies of these and location drawings must be furnished to the department.
- 8.8 The contractor may be required to repeat the test as required, should the ambient conditions at the time not given, in the opinion of the purchaser, sufficient and suitable indication of the effect and performance of the installation as a whole or of any part, as required.
- 8.9 In case of equipments being supplied by the contractor from a place outside India for which department may or may not be able to send its representative, the manufacturers test certificates (for all acceptance test) for inspection of material and witnessing the various tests at the factory as carried out by the manufacturer have to be submitted alongwith the supply of equipments. This must be witnessed by a representative of the contractor.

NOTES :

A. TEST INSTRUMENTS

1. All instruments for testing shall be provided by the Ventilation system contractor.
2. Thermometer shall have graduations of 0.1°C and shall be got calibrated from N.P.L. or any recognized test house before hand.
3. Thermometers used in the psychrometers shall have graduations of 0.2°C and shall be calibrated as at (2) above.
4. Pressure gauges shall also be got calibrated before hand from a recognized test houses.
5. Air flow rates shall measure in the supply duct using pitot tube.

SECTION – 8 - MODE OF MEASUREMENTS

1. UNIT PRICES IN THE SCHEDULE OF QUANTITIES

- 1.1 The item description in the schedule of quantities is in the form of a condensed resume. The unit price shall be held to include everything necessary to complete the work covered by this item in accordance with the specifications and drawings. The sum total of all the individual item prices shall represent the total price of the installation ready to be handed over.
- 1.2 The unit price of the various items shall include the following:
 - 1.2.1 All equipment, machinery, apparatus and materials required as well as the cost of any tests which EIC may request in addition to the tests generally required to prove quality and performance of equipment.
 - 1.2.2 All the labour required to supply and install the complete installation in accordance with the specifications/drawings.
 - 1.2.3 Use of any tools, equipment, machinery, lifting tackle, scaffolding, ladders etc. required by the contractor to carry out his work.
 - 1.2.4 All the necessary measures to prevent the transmission of vibration.
 - 1.2.5 The necessary material to isolate equipment foundations from the building structure, wherever necessary.
 - 1.2.6 Storage and insurance of all equipment apparatus and materials.
- 1.3 The contractor's unit price shall include all equipment, apparatus, material and labour indicated in the drawings and/or specifications in conjunction with the item in question, as well as all additional equipment, apparatus, material and labour usual and necessary to make in question on its own (and within the system as a whole) complete even though not specifically shown, described or otherwise referred to.
 - 1.3.1 The contractor's unit price shall also include all taxes, duties, octrois, works contract tax etc. as applicable.

2. MEASUREMENTS OF SHEET METAL DUCTS, GRILLES/ DIFFUSERS ETC.

2.1 SHEET METAL DUCTS

- 2.1.1 All duct measurements shall be taken as per actual outer duct surface area including bends, tees, reducers, collars, vanes & other fittings. Gaskets, nuts, bolts, vibration reducing pads are included in the basic duct items of the SOQ./ or as per IS code.
- 2.1.2 The unit of measurements shall be the finished sheet metal surfaces area in meters squares. No extra shall be allowed for laps and wastage.
- 2.1.3 All the guide vanes, deflectors in duct elbows, branches, grille collars quadrant dampers etc. shall be measured for actual sheet metal surface and paid for at the same rate as duct of same thickness.

- 2.1.4 The unit duct price shall include all the duct hangers and supports, exposing of concrete reinforcement for supports and making good of the same as well as any materials and labour required to complete the duct frame.

2.2 GRILLES/DIFFUSERS

All grilles/diffusers as per tender requirements shall be measured as follows.

- 2.2.1 All measurements of grilles/diffusers shall be the actual outlet size excluding the outer flanges.
- 2.2.2 The square or rectangular grills/diffusers shall be measured in sq.m.
- 2.2.3 All round diffusers shall be measured by their diameters in cm.
- 2.2.4 All linear diffusers shall be measured as per actual length in metres.

3. MEASUREMENTS OF PIPING, FITTINGS, VALVES, FABRICATED ITEMS

3.1 PIPE

- 3.1.1 All pipes shall be measured in linear meter (to the nearest cm) along the axis of the pipes and rates shall be inclusive of all fittings e.g. tees, bends, reducers, elbows etc. Deduction shall be made for valves in the line.
- 3.1.2 The rate quoted shall be inclusive of cutting holes, exposing reinforcement in wall and ceiling and floors and making good the same and inclusive of all items as specified in specification and schedule of quantities.
- 3.1.3 Rates quoted shall be inclusive of providing and fixing pads wherever specified or required by the Engineer in charge.
- 3.1.4 Flexible connections, wherever required or specified shall be measured as part of straight length of same diameter, with no additional allowance being made for providing the same.
- 3.1.5 The length of the pipe for the purpose of payment will be taken through the centerline of the pipe and all fittings (e.g. tees, bends, reducers, elbows, etc.) as through the fittings are also resumed to be pipe lengths. Nothing extra whatsoever will be paid for over and above for the fittings.

3.2 STRUCTURAL SUPPORTS

Structural supports including supports fabricated from pipe lengths for pipes shall be measured as part of pipe line and hence no separate payment will be made. Rates shall be inclusive of hoisting, cutting, jointing, welding, cutting of holes and chases in walls, slabs or floors, painting supports and other items as described in specifications, drawings and schedule of quantities or as required at site by Engineer in Charge.

4. INSULATION

- 4.1 The measurement for piping, and ducts shall be made over the bare uninsulated surface area of the metal.

4.2 PIPES, DUCTS

4.2.1 PIPES

The measurements for insulation of piping shall be made in linear metres through all valves, flanges, and fittings. Pipes/bends shall be measured along the centerline radius between tangent points. If the outer radius is R1 and the inner radius is R2. The center line radius shall be measure as $(R1+R2)/2$. Measurement of all fittings shall be measure with the running metre of pipe line as if they are also pipe lengths. Nothing extra over the above shall be payable for insulation over fittings in pipe line/routings. Fittings that connect two or more different sizes of pipe shall be measured as part of the larger size.

4.2.2 DUCTS

The measurements for insulation of ducts shall be made in actual square metres of bare uninsulated duct surface through all dampers, flanges and fittings. In case of bends the area shall be worked out by taking an average of inner and outer lengths of the bends. Measurements for the dampers, flanges, fittings shall be for the surface dimension for the connecting duct, nothing extra over the above shall be payable for insulation over dampers, flanges and fittings in duct routing.

4.3 ACCESSORIES INSULATION

4.3.2 In case of curved or irregular surfaces, measurements shall be taken along the curves.

4.3.3 The unit insulation price shall include all necessary adhesives, vapour proofing and finishing materials as well as additional labour and material required for fixing the insulation.

4.4 ROOF AND WALL INSULATION & ACOUSTIC TREATMENT

4.4.1 The unit of measurement for all underdeck roof insulation, wall insulation, wall/roof acoustic panel shall be the uninsulated area of walls, roofs, to be treated, in square meters.

4.4.2 The insulation, acoustic panels shall include cost of battens, supports, adhesives, vapour proofing, finished tiles/boards/sheets as well as additional labour and material required for completing the work.

APPENDIX – II
TECHNICAL DATA TO BE FURNISHED BY THE TENDERER FOR THE VRF/ VRV
SYSTEM

Tenderers are required to separately fill the Data for each rating of the equipment offered.

1. Ventilation Fan

- 1.1 Manufacturer
- 1.2 Model No.
- 1.3 Type of fan
- 1.4 Overall dimension (mm)
- 1.5 Air quantity CFM
- 1.6 Static pressure (mm wc)
- 1.7 Speed RPM
- 1.8 Motor Horse Power
- 1.9 Method of Starting
- 1.10 Class of insulation

APPENDIX – III

SUBMITTALS TO THE MADE BY THE CONTRACTOR DURING THE EXECUTION OF WORK

1. Daily Progress report stating number of men employed under each trading, Equipment at site etc.
2. Weekly/fortnightly progressing report showing progress against programme.
3. Programme of work for the forthcoming week.
4. Labour and Equipment Deployed at site - Programmed
Requirement vs actual deployed - weekly
5. Updated approved monthly PER CHART along with monthly progress chart - Weekly.
6. Construction materials by Contractor : Status and mobilization programme -
Fortnightly.
7. Progress Photographs - Fortnightly.
8. Value of work anticipated to be done in the forth-coming including value of any
materials and equipment of large value - monthly